



NASA Rockets

All of NASA's space shuttles must go through New Orleans, but do you know why they must pass through this ritual? <http://dgit.in/NasaRockets>



SLS usage questioned

Space launch system missiles are under review, as new comments suggesting NASA shouldn't launch these expensive missiles <http://dgit.in/SLSCritic>

Space Age

had to cut short their mission and were transferred back to Soyuz 11 for re-entry to Earth. Tragically, a pressure relief valve on the spacecraft malfunctioned during re-entry causing a loss in cabin pressure and the death of the brave cosmonauts.

America's entry into the space station race

Seeing the success of the Salyut program, the United States was not one to hand in the towel, especially when it meant conceding to the Russians. Unfortunately, funding for space programs were at a premium, especially after the severe dent the Apollo program caused in America's fiscal budget. As the program began to lose steam in the early 70's, NASA (the National Aeronautics and Space Administration) started the Apollo Applications Program, which aimed to launch unused hardware (leftover from the Moon missions) into space for research and exploration.

It was in this program that rocket engineer Wernher von Braun conceived the idea of building a space station from an unused rocket stage. Despite reduced funding, NASA scientist tested and tweaked various prototypes until they landed on the final design of what would be the United State's first ever orbiting space station: Skylab.

Houston, we have a problem

The day of the launch of Skylab, May 14 1973, is a day that will go down in the history books of NASA as one where Murphy's Law reigned supreme. The myriad of technical issues faced by the astronauts and ground crew of Skylab is almost legendary. It all began 63 seconds into its

THE FIRST SPACE TOURISTS

As the Soviet Space Program began getting pressed for funds, they thought up the ingenious idea of space tourism to help inject some essential cash flow for further Space Exploration. Japanese TV channel TBS were the first to strike a deal with the Russians, which involved flying reporter Toyohiro Akiyama to Mir, a privilege for which they reportedly paid between \$12 to \$37 million. Another famous space tourist to Mir was chemist and Mars chocolate company employee Helen Sharman, who also has the honour of being the first ever Briton in space. Sharman was selected through Project Juno, a program conceived as cooperative arrangement between the Soviet Space Program and a group of British companies.

launch, when the earth's atmosphere tore off the ship's meteoroid shield taking with it two primary solar panels and damaging the third one, thus depriving Skylab most of its electrical power. The situation only seemed to get worse as additional damage to the ship's antenna confirmed that the ship's communications system had severely malfunctioned.

The ground crew at NASA's Marshall Space Flight Center rushed to stabilize the situation by putting the station at an altitude that would minimize overheating. Crew members were forced to conduct emergency repair procedures that included manually deploying a solar array, erecting a sunshade to avoid overheating and con-

ducting an improvised docking procedure after a failed initial docking attempt. While definitely looked at as a sour note in the history of American Space Exploration, Skylab's botched launch also proved that fixing a damaged space station while still in orbit was not an impossible task.

During its orbital lifetime, Skylab's crews were able to conduct a number of crucial scientific experiments, including confirming the existence of coronal holes in the Sun and using sensors to recorded data in the visible, infrared, and microwave spectral regions of earth. Skylab's last crew departed the space station in February 1974, leaving it in unmanned orbit. Despite NASA's repeated plans to send additional crews, budgetary restrictions and the advent of additional space programs meant focus had to be shifted elsewhere.

Russia's re-entry into the space station game

The United State's withdrawal from manned space station programs left the door open for the Russians, who were itching to discover the next technological breakthrough in space exploration. Enter the Mir space station. Famous for setting numerous space-related records, Mir will be best remembered for two truly pioneering feats. It was the first space station that boasted a modular design and was constructed in space by engineers who had to assemble individually launched modules – starting with its core module that was launched in 1986.

Secondly, instead of following traditional protocol that involved bringing home all cosmonauts posted on the station and subsequently launching a new crew, the Russians decided instead to conduct "changing of guard" that would take place on board the outpost itself. This gave Mir its famous moniker of being Earth's first ever permanently manned extraterrestrial outpost. However, this so-called permanent status was severely tested in April 1987, when Mir cosmonaut Aleksandr Laveikin was diagnosed with potential cardiovascular problem and was in danger of being shuttled back to earth. Russia was steadfast in their resolve, and instead of putting an end to long-standing expedition record, they sent within three months a replacement



Full scale model of the Salyut 7 with two docked spacecrafts